

SATHWIK VADALA

AI / Full-Stack Software Engineer

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LinkedIn | GitHub | Portfolio

Professional Summary

AI / Full-Stack Software Engineer with 2+ years of professional experience building AI-enabled and business applications using **Python, FastAPI, React, Next.js, TypeScript, Node.js, PostgreSQL, REST APIs, and Generative AI**. Experienced in developing LLM-powered applications, AI copilots, structured AI workflows, document processing, prompt engineering, API integrations, authentication, and full-stack application architecture.

Hands-on experience with **RAG, semantic search, embeddings, vector retrieval, AI agents, tool calling, MCP concepts, LangChain workflows, prompt chaining, structured outputs, and LLM application design**. Strong foundation in backend API development, database integration, debugging, testing, deployment, and AI-assisted software development.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, HTML, CSS

Frontend: React.js, Next.js, Tailwind CSS, Responsive Web Applications, Progressive Web Applications

Backend: FastAPI, Node.js, Express.js, REST APIs, Async Programming, API Integration, Microservices Concepts

Generative AI: Generative AI, Large Language Models, Gemini, LLM Integration, Prompt Engineering, Structured Outputs, AI Copilots, AI Agents, Agentic AI

RAG & AI Systems: RAG, Semantic Search, Embeddings, Vector Search, Knowledge Grounding, Document Processing, Tool Calling, Function Calling

AI Frameworks & Protocols: LangChain, LangGraph, Model Context Protocol (MCP), MCP Tool Integration, Agent Workflows

Databases & Authentication: PostgreSQL, Prisma ORM, SQL, IndexedDB, OAuth, JWT, Google OAuth, Auth.js

Cloud & Tools: Vercel, AWS Concepts, Git, GitHub, CI/CD Concepts, Figma, AI-Assisted Development, Modern IDEs

Professional Experience

GenAILakes Pvt. Ltd.

Full Stack Developer / AI Software Engineer

Aug 2024 – Present

Hyderabad, Telangana

- Develop and maintain full-stack applications using **Python, FastAPI, React, Next.js, TypeScript, Node.js, and PostgreSQL**, covering frontend interfaces, backend services, REST APIs, databases, authentication, and external integrations.
- Develop RESTful APIs and integrate application services with databases, third-party APIs, authentication providers, and AI services to create end-to-end application workflows.
- Build and integrate **Generative AI and LLM-powered functionality** for information processing, document analysis, recommendations, automation, and user-assistance workflows.
- Apply **prompt engineering, progressive prompting, prompt chaining, structured prompting, and output validation** to improve reliability and consistency of LLM-powered workflows.
- Work with AI application patterns including **RAG, semantic retrieval, embeddings, vector search, tool calling, and agent workflows** for context-aware LLM applications.
- Develop hands-on knowledge of **Model Context Protocol (MCP)** and MCP-enabled tool integration patterns for connecting AI systems with APIs, developer tools, databases, and external services.
- Design reusable frontend and backend components with emphasis on maintainability, validation, error handling, API consistency, and development efficiency.
- Use Figma and Figma-to-code workflows to translate interface designs and component specifications into responsive React and Next.js applications.
- Perform application testing, debugging, root-cause analysis, and issue resolution across frontend, backend, API, database, and deployment environments.

Selected AI Projects

AI Career Copilot

Next.js, TypeScript, Python, FastAPI, PostgreSQL, Prisma, Gemini, Auth.js

- Built an AI-powered career platform that analyzes resumes and job descriptions to generate ATS-oriented scores, identify matched and missing skills, and provide career recommendations.
- Designed an end-to-end LLM workflow covering resume ingestion, job-description processing, context preparation, AI

analysis, structured outputs, and user-facing recommendations.

- Developed **Python / FastAPI REST APIs** for resume processing, job analysis, authentication, user workflows, and AI-powered operations.
- Integrated Gemini-based LLM capabilities using **prompt engineering, structured outputs, and validation** for consistent AI-generated results.
- Designed PostgreSQL data models using Prisma ORM and developed reusable Next.js and TypeScript application components.
- Implemented Google OAuth and JWT-based authentication patterns for protected application workflows.

SpendTrack AI – AI Personal Finance Platform *React, TypeScript, Node.js, Express, Prisma, PostgreSQL, Gemini AI*

- Designed an AI-powered personal finance platform combining transaction management, budgeting, planning, secure data storage, financial intelligence, receipt processing, and an AI Copilot.
- Designed an **AI Copilot workflow** for answering natural-language questions using application transaction, budget, and financial context.
- Designed receipt-processing workflows using Generative AI to transform unstructured receipt information into structured application-ready data.
- Designed workflow orchestration connecting Transactions, Budget, Planner, Vault, Financial Intelligence, Receipt AI, and AI Copilot components.
- Developed backend services and database integrations using Node.js, Express, Prisma, and PostgreSQL with structured API workflows.
- Designed offline-first functionality using IndexedDB and Service Workers and applied WebCrypto / AES-GCM concepts for secure local data storage.

Education

Osmania University

Dec 2023

Master of Computer Applications (MCA)

Satavahana University

Dec 2021

Bachelor's Degree

Training

ISEA Phase-III Linux & Systems Security Bootcamp – IIT Hyderabad

Hands-on training covering Linux systems, system security, networking, and security fundamentals.